

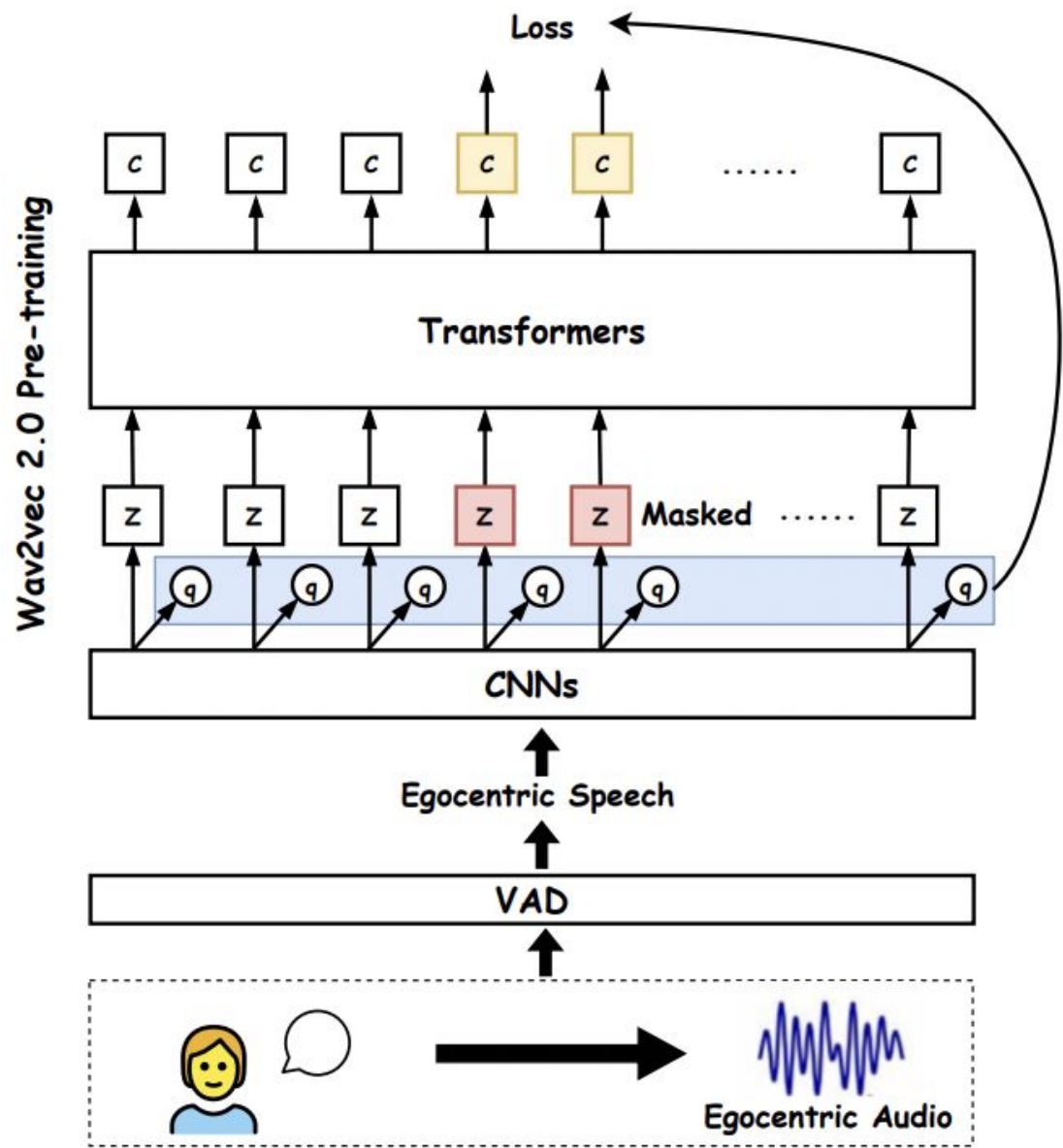
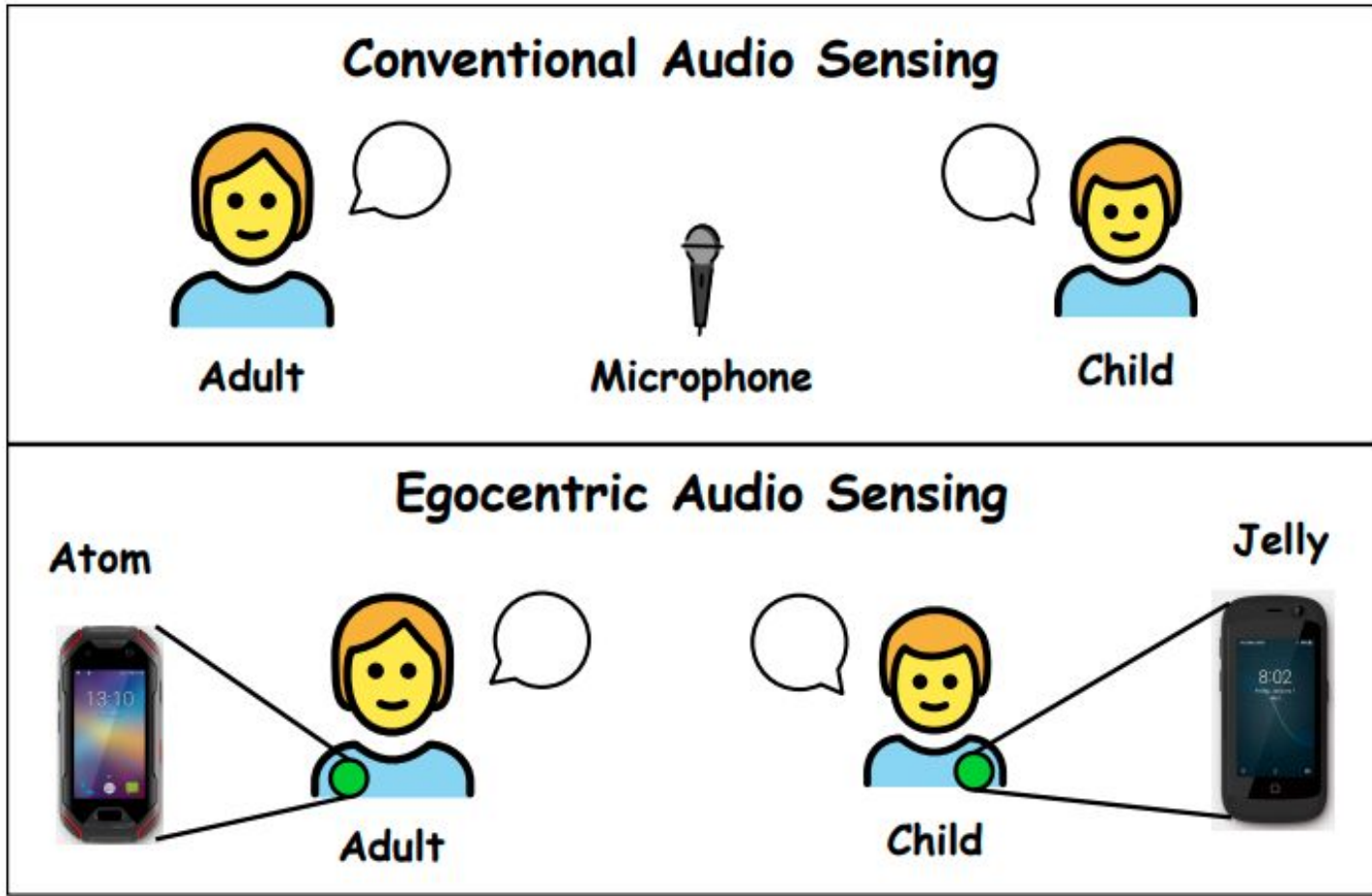
Human-centered Speech Computing across Life Span

Tiantian Feng, Shrikanth Narayanan and many SAILers
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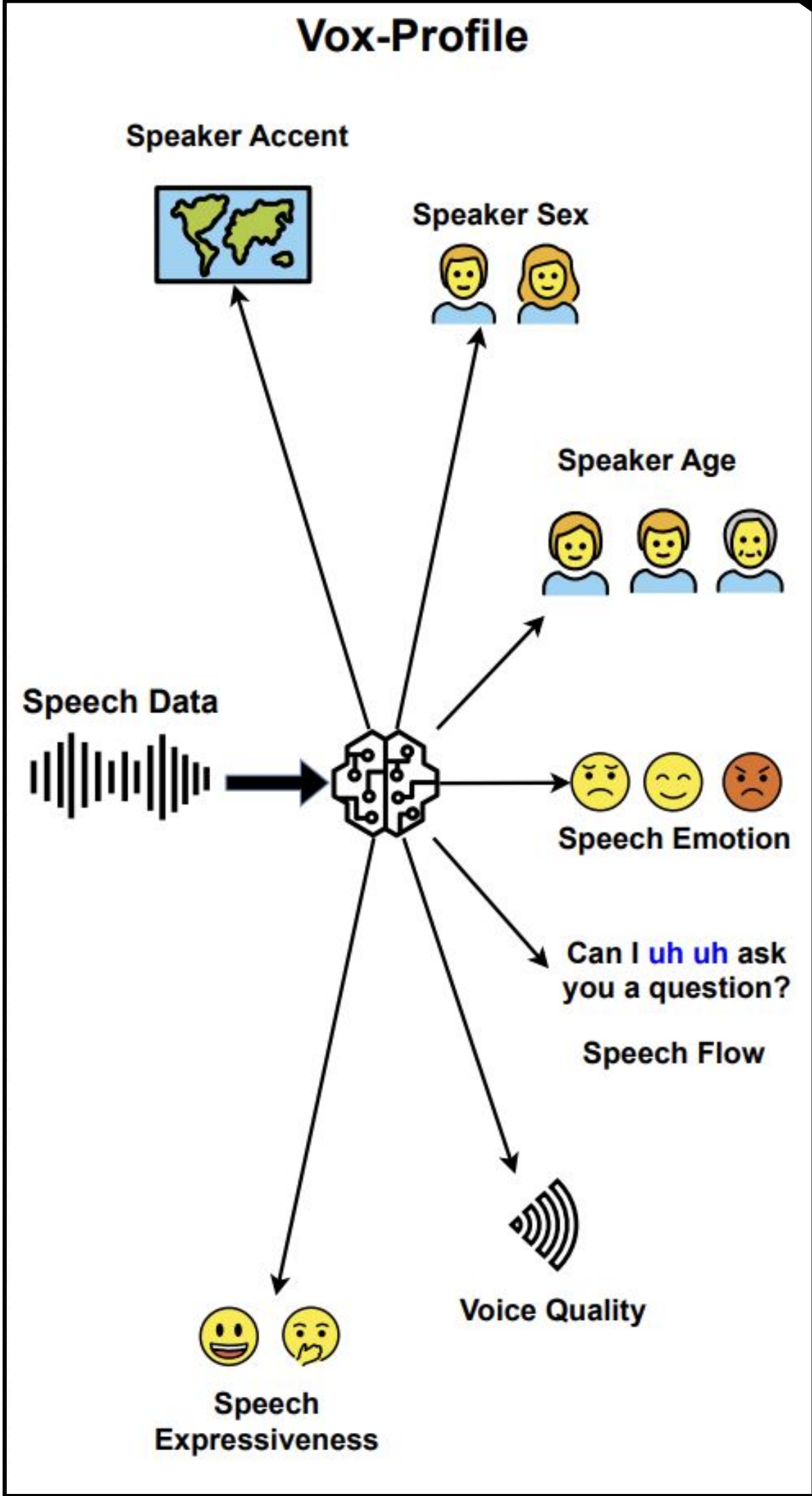


Child Speech Modeling

Child

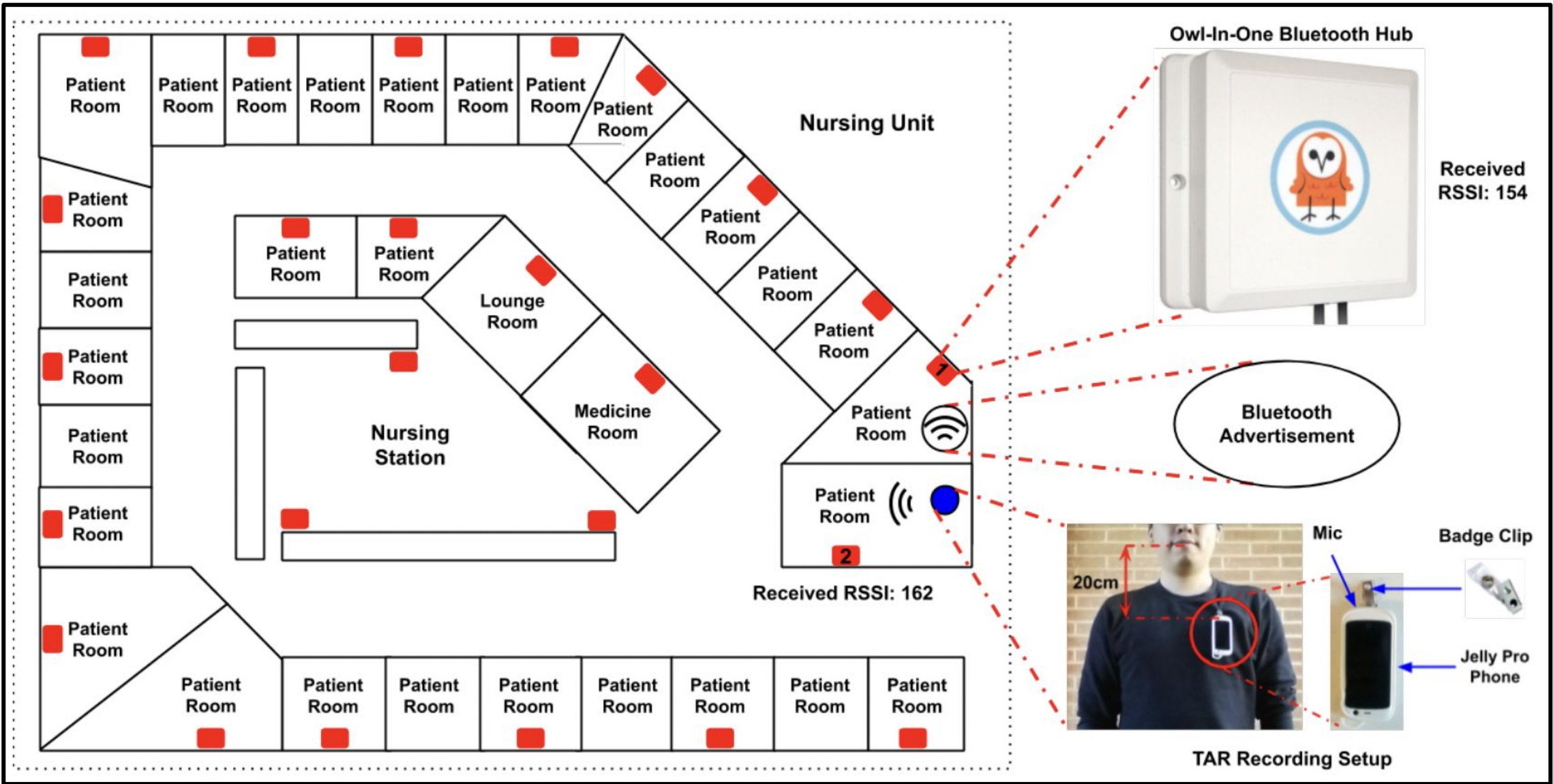


Test Set	Backbone	Acc	F1	Recall	Specificity
Jelly-Child	W2V	84.0	0.801	77.2	87.8
	Ego4D W2V	85.4	0.812	74.2	90.7
Atom-Exam	W2V	84.8	0.810	81.3	87.6
	Ego4D W2V	87.5	0.842	80.9	90.8



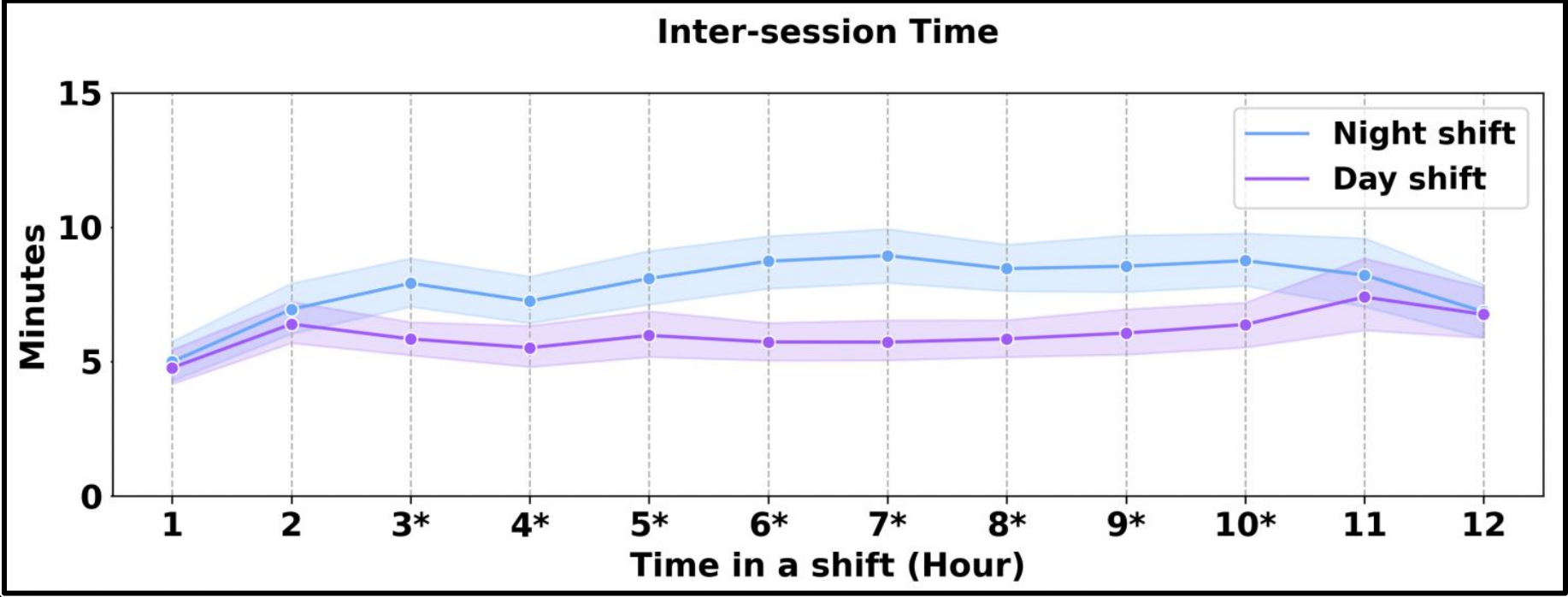
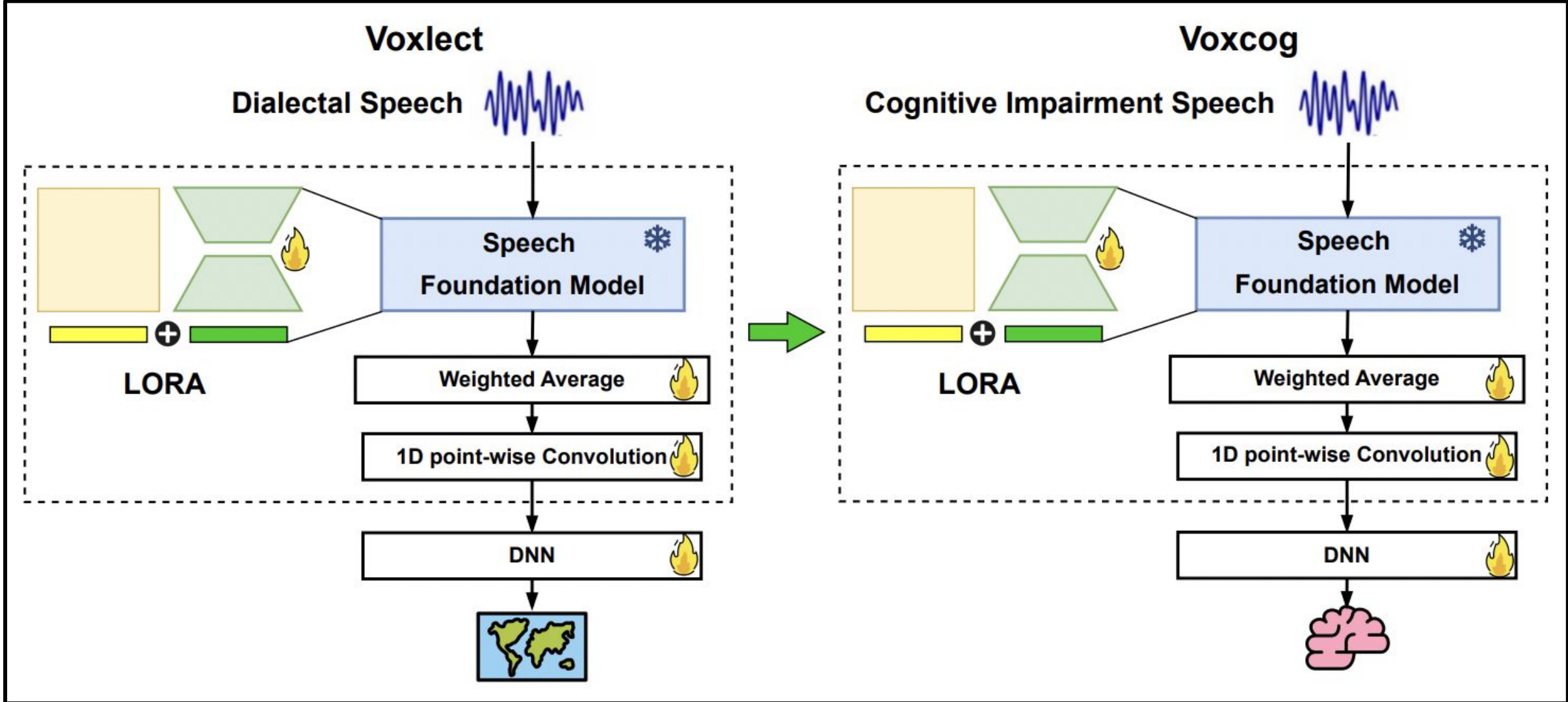
Adulthood

Workplace Behavior



Speaker Modeling

Cognitive Impairment Detection



Elder

Dataset	Model	Macro-F1	Accuracy
2020-ADReSS	Challenge Baseline(Luz et al. (2020))	0.745	75.00
	INESC-ID(Pompili et al. (2020))	0.808	81.25
	JHU x-vector(Pappagari et al. (2020))	0.745	75.00
	RMIT System(Syed et al. (2020))	-	<u>85.42</u>
	VoxCog (Ours)	0.875	87.50